Johns Hopkins University

Real-Time FBG-Based Stylet Shape Sensing for Cervical Cancer Treatment

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Chapter 1

Introduction

1.1 Project Goal

This project aims to develop a **real-time shape sensing system using Fiber Bragg Grating (FBG) technology** to guide catheter insertion during brachytherapy treatment for locally advanced cervical cancer.

1.2 Relevant Terminology

Technical terms:

1. Fiber Bragg Grating (FBG): An optical sensor which can be used to detect changes in strain throughout a fiber from a reflected optical wave (Lezcano, et al, 2022).
2. Optical fiber: A fiber that transmits light signals and contains FBGs at known locations for strain detection. (Lezcano, et al, 2022).
3. Multicore fiber (MCF): A collection of optical fibers with FBGs arranged radially. (Lezcano, et al, 2022).

Medical terms:

1. Brachytherapy: A cancer treatment that delivers radioactive seeds through catheters to target volumes (in this case, cervical cancer tumors) ("Brachytherapy to treat cancer", 2019).

1.3 Background

Cervical cancer is currently the third most common cancer among women worldwide, with 13,360 new cases and 4,320 deaths expected this year in the United States ("Cancer Stat Facts", 2025). Locally advanced cases of cervical cancer occur when the cancer has grown beyond the cervix but has not fully reached distant organs. Currently, brachytherapy is the standard of care for these cases (Banerjee, Kamrava, 2014). Brachytherapy facilitates the delivery of high-dose radioactive seeds directly to tumors with minimal exposure to surrounding tissues and organs ("Brachytherapy to treat cancer", 2019). Radioactive seeds are delivered through catheters, which must be placed in the tumor to avoid damage to non-cancerous tissues, as shown in Figure 1.1.

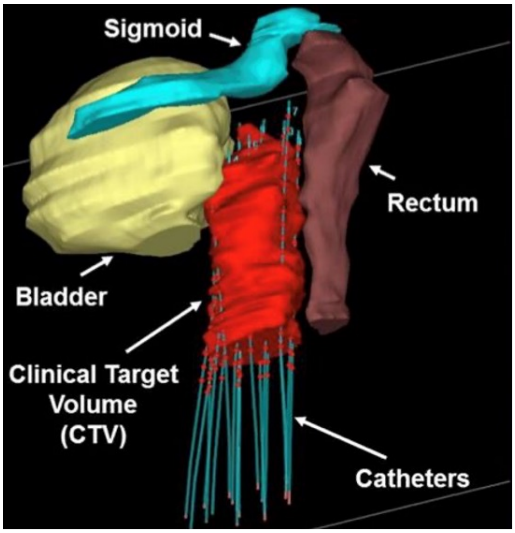

Figure 1.1: Catheters placed in the CTV (tumor). Photo from NIH Grant #1R01EB036015-01.

Current catheter placement techniques use medical imaging (MRI, ultrasound, or CT) for guidance (Manea, et al., 2024). The typical workflow for catheter insertion involves obtaining a preoperative MRI or CT of the tumor to plan the trajectory for each catheter. Then, the insertion of the catheters is performed in the operating room under ultrasound guidance. Finally, there may be a postoperative MRI or CT to verify the placement of the catheters. If the catheters are misplaced, a radiation physicist will lower or stop the delivery of radiation through that catheter to avoid damage to surrounding organs. The catheters and applicator must stay in place for the duration of the procedure, which can be over 24 hours long.

This workflow is complex, as it involves multiple types of imaging which involves movement of the patient and takes time. On top of this, the full brachytherapy treatment can take over 24 hours, so it is likely that the patient may move during this time and catheters may become misguided. The development of the FBG-based shape sensing stylet aims to simplify this workflow.

1.4 Significance

Clinicians need a way to perform adaptive catheter placement for brachytherapy applications using a simplified workflow. Real-time tracking of catheter trajectory can be used as feedback for adaptive catheter placement in cervical cancer brachytherapy approaches, simplifying workflow, shortening procedure time, and limiting radiation exposure.

1.5 Specific Goals

- Calibration procedure documentation and resulting dataset (<0.5mm error)
- Gel phantom experiment procedure documentation and resulting dataset (<1.2mm error)
- Real tissue experiment process documentation and resulting dataset (<2mm error)

Chapter 2

Technical Summary

2.1 Reconstruction Model

Prior literature has shown that raw FBG wavelength data can be used to determine the curvature of a rod in each sensor location according to the linear relationship (Lezcano, et al, 2022):

$$\begin{bmatrix} \kappa_{xz} \\ \kappa_{yz} \end{bmatrix} = C \begin{bmatrix} \Delta\lambda_1 \\ \Delta\lambda_2 \\ \vdots \\ \Delta\lambda_7 \end{bmatrix}$$

Our goal is to determine the calibration matrix, C. The first step is to define the reference wavelength, λ_R. This process will be described in the calibration section. Then, the stylet will be bent to a known, constant curvature in the xz and yz planes and FBG wavelength data will be collected. From this data, we define the wavelength shifts as $\Delta\lambda = \lambda - \lambda_R$, where λ is the wavelength at the sensor location for the known constant curvature. Then, by collecting data for multiple known curvatures, we can find the calibration matrix by using the pseudoinverse and weighted least squares regression (Lezcano, et al, 2022):

$$\left(\begin{bmatrix} \Delta\lambda_1 \\ \Delta\lambda_2 \\ \vdots \\ \Delta\lambda_7 \end{bmatrix}^T W \begin{bmatrix} \Delta\lambda_1 \\ \Delta\lambda_2 \\ \vdots \\ \Delta\lambda_7 \end{bmatrix} \right) C = \left(\begin{bmatrix} \Delta\lambda_1 \\ \Delta\lambda_2 \\ \vdots \\ \Delta\lambda_7 \end{bmatrix}^T W \begin{bmatrix} \kappa_{xz} \\ \kappa_{yz} \end{bmatrix}^T \right)$$

The final step is to compute reliability weightings for each of the active areas. This is done by calculating the mean square error between the computed curvature value by the FBG sensors and the known ground truth curvatures. Then, each mean square error is inverted, such that larger errors lead to smaller weightings. Finally, the weightings are normalized so that the final curvature can be computed by the weighted sum from each active error.

Once the sensor model has been defined by the calibration matrices, the shape of the stylet for constant curvatures can be computed from the raw wavelength data. This will be done for the validation curvatures to confirm that the calibration matrices found are accurate. The raw wavelength shifts from each sensor location are multiplied by the calibration matrix to compute the curvature value at that location. Since the curvature is constant for this insertion, the weighted sum from all the active areas is found to define the current shape. Integration of this curvature across the entire length of the stylet results in the stylet shape.

When the curvature is not constant and the stylet is inserted into a tissue, the Lie-group theoretic model will be used (Lezcano, et al, 2022). The process involves using the FBG wavelength data to optimize a cost function using interior-point nonlinear optimization to find the optimal parameters, then solving the Euler-Poincaré equation using these optimal parameters and the rotation matrix defined by the Lie-group theoretic model (Lezcano, et al, 2022). Finally, Simpson vector integration of the rotation matrix yields the positions of the stylet allowing for the complete shape reconstruction.

2.2 Calibration & Validation

The goal of the calibration experiment was to determine the calibration matrices that convert the raw wavelength data from each FBG sensor to the curvature induced at that sensor. The mathematical relationship was discussed above in Section 2.1.1. The calibration and validation processes were executed by using linear stage to move the jigs and rotary stage to rotate the needle. 5 insertions were performed for zero degree rotation for each curvature on the jig, then the needle was rotated to 90 degrees and the 5 insertions were repeated for each curve. The rotations performed were 0 and 90 degrees to ensure functionality in the x-y and y-z planes before attempting intermediate planes at other rotation angles.

2.2.1 FBG Interrogator Setup

For each of the 14 active areas on each of the 7 fibers, a wavelength will be read using the interrogator. The FemtoSense GUI was used to create peak sensors for each of the 98 sensors. The mean and standard deviation of the peak value of a "sweep" of the interrogator are reported around every 2 seconds by the interrogator. We determined that for optimal results, a low standard deviation is critical. This required careful selection of the sensor bounds, threshold value for the peak to be above, and the number of fit points above the threshold required.

2.2.2 Linear Stage & Rotary Setup

Linear stage only translated backwards and forwards and was designed to prevent sideways translation. The rotary stage was designed such that it only translated sideways and locked in position every 10mm - the exact distance between our jig curves. The rotary stage and linear stage were arranged to be exactly perpendicular to each other.Thus minimal error was introduced by these components.

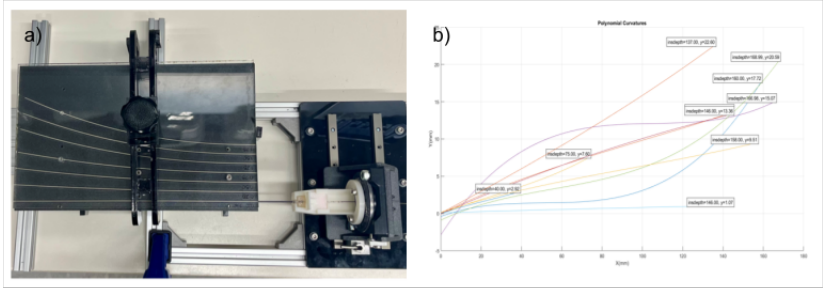

Figure 2.2: (a) Calibration jig and stages setup; (b) 10 Clinical needle curvatures characterized by insertion depth and deviation from center

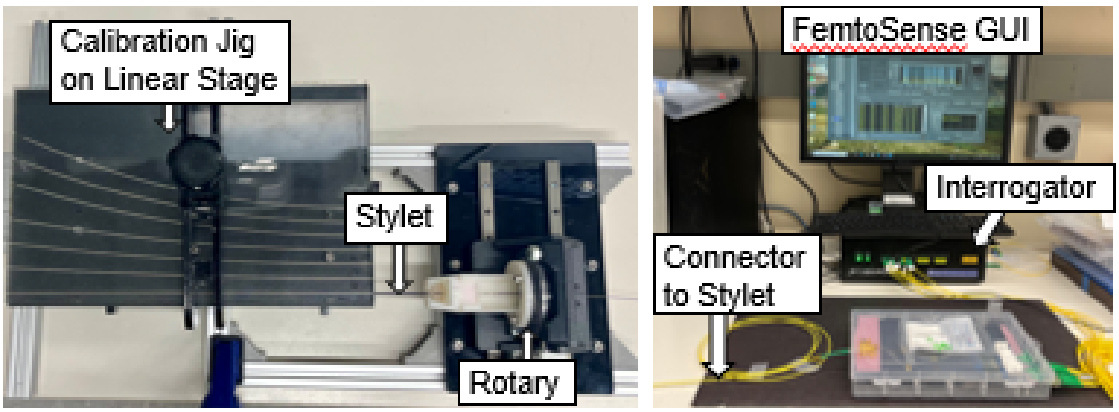

Figure 2.1: System setup for calibration test

2.2.3 Jig Design

Much of our work for the calibration experiment involved designing calibration jigs for clinically relevant ground-truth curvatures. We received procedural needle curvature data from our clinical partners at Brigham and Women's Hospital, with more than 100 needle placements extracted from MRI scans across 5 brachytherapy patient procedures and curve fit to third-order polynomial functions. We picked curvatures that were clinically relevant ($< 3.17\ m^{-1}$) but still small enough to prevent permanent needle deformation. The final calibration jig contained five non-zero curvatures (0.25, 0.50, 0.75, 1.25, 1.50 m^{-1}) and one zero curvature to use as reference. The final validation jig contained four non-zero curvatures (0.33, 0.66, 1.0, 1.33 m^{-1}).

2.3 Gel Phantom Validation

During the experiment, the needle was manually inserted into the gel phantom to a maximum depth of 220 mm. As the needle deformed within the phantom, the embedded FBG sensors captured strain data, which was then used to reconstruct the needle's 3D shape. A stereo camera system was used to visually track the needle tip position to independently validate the accuracy of the reconstructed shape.

2.3.1 Stereo Camera Setup

3D reconstruction

To obtain ground truth data of the needle shape, a stereo vision system was employed. The stereo camera setup was calibrated in advance to eliminate lens distortion and to extract both intrinsic and extrinsic parameters of the cameras. The reconstruction process involves the following steps:

1. Needle Segmentation: A reference image of the scene without the needle is subtracted from the image with the inserted needle. This highlights the needle, allowing for accurate segmentation in both stereo images.

2. Centerline Extraction: The needle is segmented in each stereo image, and its centerline is extracted using standard image processing techniques shown in Figure 2.2.a.

3. Image Rectification and Spline Fitting: Stereo rectification is performed to align the image pairs along epipolar lines. The extracted centerlines are then smoothed and interpolated using second-order B-splines to improve stability and curve continuity. The result is shown in Figure 2.2.b.

4. Stereo Matching: Template matching is conducted along the epipolar lines at each point of the centerline, using images upscaled by a factor of 2.5 to increase resolution. The resulting matched points are again smoothed and interpolated using B-splines. The figure is shown in 2.2.c and 2.2.d.

5. Triangulation: The interpolated stereo correspondences are used to perform 3-D triangulation, reconstructing the needle's spatial shape.

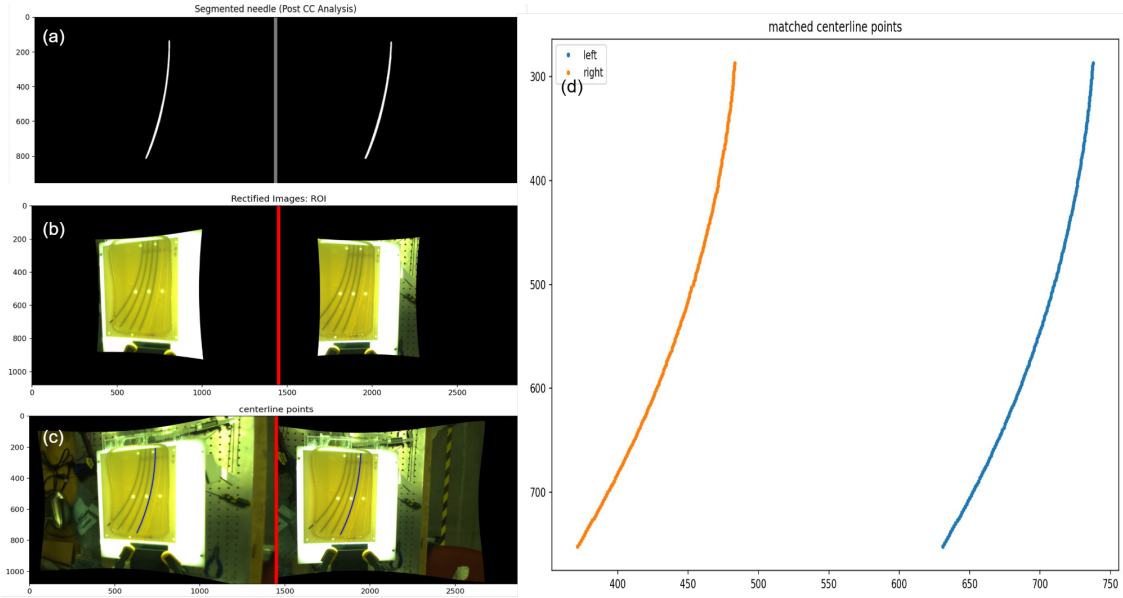

Figure 2.3: image process and reconstruction of 2.0 1/m tube under stereo camera: (a) Segmented needle; (b) Retiflied image; (c) centerline points of the curve; (d) matched centerline points

Calibration of stereo camera

Stereo camera calibration was performed using a 9×11 checkerboard grid with a gel layer applied over its surface. The presence of gel simulates the optical interface conditions encountered during actual imaging through tissuemimicking phantoms. We use the ROS camera_calibration package, resulting in the generation of intrinsic and extrinsic calibration parameters stored in an ost file. This file includes each camera's focal length, principal point, distortion coefficients, and the stereo baseline.

To validate the accuracy of the stereo camera system, tubes with known curvatures of 2.0 1/m and 2.5 1/m were placed into a custom-designed jig. This jig held the tubes in place beneath the gel phantom. Stereo images were captured under this setup, and the corresponding 3D reconstructions of the tubes were performed using the previously calibrated camera parameters. The figure 2.3 (a) shows the image with 2.5 1/m curvature tube with 220 mm length. The figure 2.3 (b) shows the image with 2.0 1/m curvature tube with 220 mm length. The figure 2.4 shows the 3D line reconstructed from stereo camera, the tip error for 2.5 1/m tube is 0.88 mm and 2.0 1/m tube is 0.76 1/m.

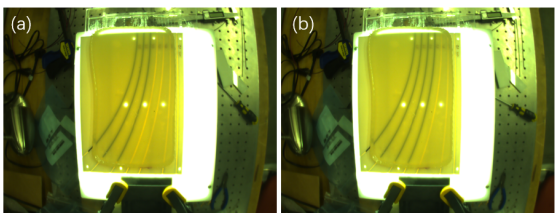

Figure 2.4: (a) image with 2.5 curvature (b) image with 2.0 curvature

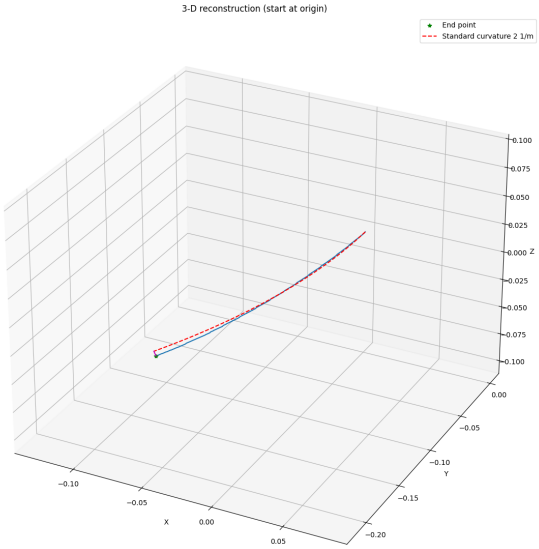

Figure 2.5: Stereo camera 3D reconstruction curvature and standard curvature

2.3.2 Gel Phantom & Container Setup

Isotropic gel phantoms are widely used in biomedical imaging to simulate the mechanical and acoustic properties of biological tissues. In this study, we constructed phantoms using plastisol materials of varying stiffness to emulate both soft and stiff tissue environments. A custom laser-cut container was fabricated to ensure precise phantom geometry and reproducibility in experimental conditions. The phantoms were fabricated using plastisol (M-F Manufacturing Company, TX, US). Two types of mechanical stiffness were targeted:

1. Stiff tissue simulation: A mixture of liquid plastisol and plastic hardener in a 4:1 volume ratio was used. This formulation yields higher mechanical stiffness, approximating fibrotic or tumorous tissue behavior.

2. Soft tissue simulation: Pure liquid plastisol was used without any added hardener, resulting in a compliant phantom that mimics healthy soft tissue.

All mixtures were degassed and cast under controlled temperature conditions to ensure uniformity and to prevent bubble formation during curing. The detail procedure is shown in the gel phantom cooking document. In the gel phantom validation, we use the stiff tissue as an inserted object to test the needle curvature.

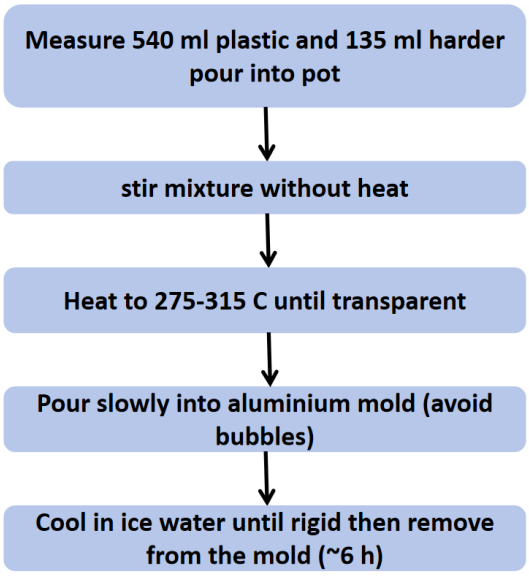

Figure 2.6: gel phantom cooking documentation

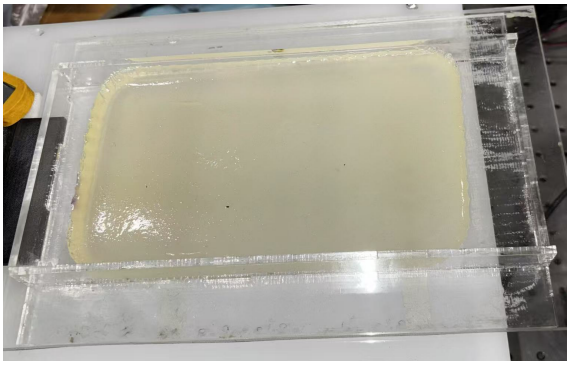

Figure 2.7: gel phantom box setup

2.3.3 Insertion Robot Setup

In our current experimental setup, the needle is inserted to a fixed depth of 150 mm. To ensure consistent insertion, the rotary stage of the insertion robot is manually adjusted to a calibrated position that allows the needle to penetrate the gel phantom to the target depth. This configuration guarantees that the entire inserted segment of the needle lies within the stereo camera's field of view and can be accurately extracted for 3D reconstruction.

Chapter 3

Results

As mentioned in the specific goals section, we will be using two error metrics to assess the stylet reconstruction. The first is tip error (TE):

$$TE = ||r_1(s_N) - r_2(s_N)||$$

r_1 is the ground truth position matrix and r_2 is the calculated position matrix. s_i is the position along the stylet and N is the total number of inserted active areas. The second is root-mean-square error (RMSE):

$$RMSE = \sqrt{\frac{1}{N}\Sigma_{i=1}^{N}||r_1(s_i) - r_2(s_i)||^2}$$

3.1 Calibration & Validation Results

Our goal for the reconstruction of the four validation curvatures was for both RMSE and TE to be less than or equal to 0.5mm. We achieved an average TE of 1.05 ± 0.277mm and an average RMSE of 0.472 ± 0.124mm. The results for each of the curvatures validated in both the XZ and YZ planes are shown in figures 3.1 and 3.2.

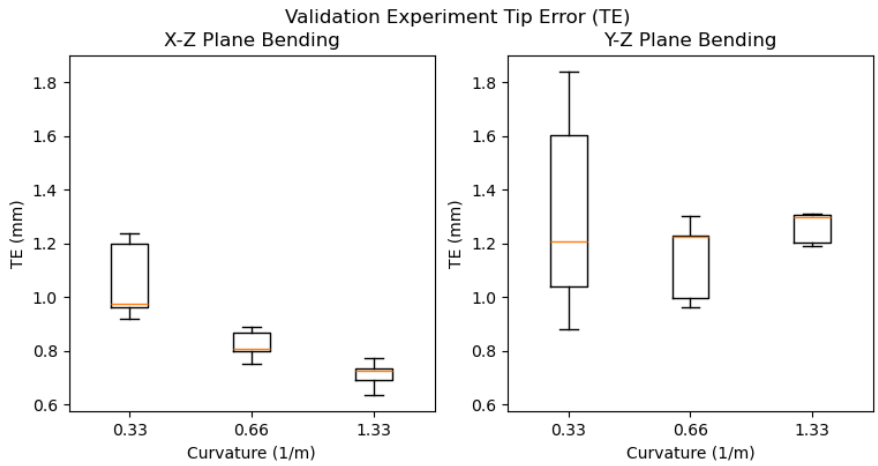

Figure 3.1: Tip Error for Validation Curvature Insertions

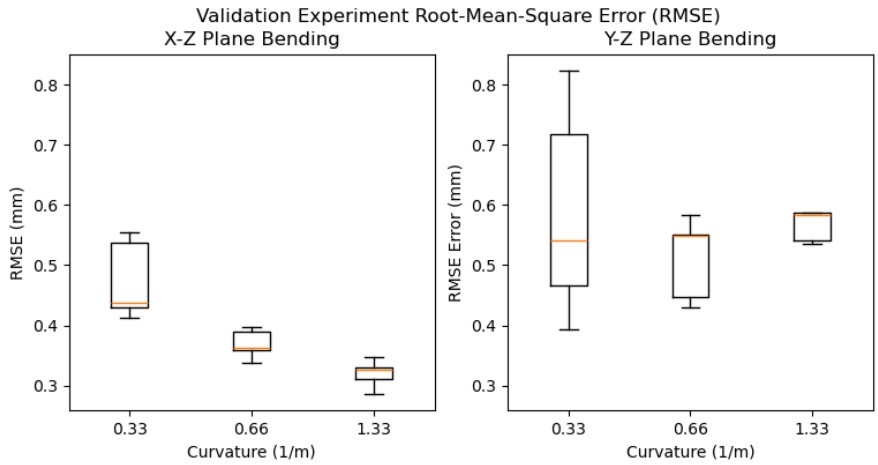

Figure 3.2: RMSE Error for Validation Curvature Insertions

As mentioned in the reconstruction model description in the technical summary, each sensor location had a reliability weighting. Figure 3.3 shows the curvature errors for each active area across all the validation curvature trials, as well as the associated weighting for that active area.

Active Area	Mean Curvature Error (1/m)	Reliability Weight	Active Area	Mean Curvature Error (1/m)	Reliability Weight
3	0.0863	0.024	9	0.0392	0.087
4	0.0302	0.079	10	0.0346	0.14
5	0.0442	0.083	11	0.0358	0.072
6	0.0640	0.11	12	0.0457	0.071
7	0.0344	0.14	13	0.0425	0.038
8	0.0509	0.13	14	0.0955	0.019

Figure 3.3: Mean curvature error from validation curvatures and associated active area reliability weighting. Note that active areas 1 and 2 were not included, as they were not inside of the jigs during the tests.

3.1.1 Discussion

We were able to meet our goals of achieving calibration and validation, but unable to achieve the required <0.50mm tip error. Rather we were able to achieve an average TE of 1.05 ± 0.277mm and an average RMSE of 0.472 ± 0.124mm. Overall the data in the X-Z plane, zero degree rotation, was generally less varied across both the tip error and RMSE than in the Y-Z plane, ninety degree rotations. This may be due to potential internal twist of the fibers induced by the rotation. In the X-Z plane there was lowest tip error and RMSE at the 1.33 1/mm curvature, and in the Y-Z plane there was lowest tip error and RMSE at the 0.33 1/mm curvature. Generally, we expected to have smallest errors at lower curvatures (Lezcano et al., 2022). A source of discrepancy could be due to an inherent bend to the needle itself; we used to needle several times and it developed a slight bend over time, specifically in the X-Z plane. This may have impacted results, and thus next steps include replacing the nitinol needle tube itself. Active areas are ordered such that 14 is near the tip of the needle and 1 is near the base. Active areas 7 and 10 were weighted the most in the algorithm, but active areas 4 and 7 display the lowest mean curvature errors. The lowest weighted active area was 14, and it also had the highest curvature error. This was expected, considering the aforementioned needle

bend, as the bend impacted the tip of the needle the most.

Chapter 4

Management Summary

4.1 Planned Project Deliverables

Minimum:

- Reconstruction software and user-guide documentation
- Built 300mm multi-core stylet
- Calibration procedure documentation & Calibration and validation dataset (<0.5mm TE and RMSE)

Expected:

- Stereo camera reconstruction software and user-guide documentation
- Gel phantom experiment procedure documentation and resulting dataset (<1.2mm TE and RMSE)

Maximum:

- CT reconstruction reconstruction software and user-guide documentation
- CT experiment procedure documentation and resulting dataset (<2mm TE and RMSE)

4.2 Accomplished Project Deliverables

From the minimum, we fully completed the following deliverables:

- Constant curvature reconstruction software and user-guide documentation
- Built 300mm multi-core stylet
- Calibration procedure documentation

We partially completed the calibration and validation dataset with <1.05mm tip error and <.472mm RMSE, as discussed in the Results section. The main setback for our project was the late arrival of the stylet (not delivered until March 23, and not usable due to thickness of nitinol tubing being too thick until April 7). We did as much as possible to prepare before the arrival of the needle, but unfortunately, it was not possible to overcome the timeline delays at the beginning of the semester and get to the error requirements. We discuss our future plans for resolving this issue in the following section.

4.3 Future Work

4.3.1 Jig-based Validation

In this work we rely on 2D c-shaped curvatures for both validation and calibration. However, in reality, the needle trajectory is rarely as well defined. Thus, future steps include incorporating s-shaped jigs as well as 3D jigs for validation to better evaluate and more closely mimic clinical needle trajectory patterns. Expanding upon this, we could better make use of the clinical needle trajectory dataset provided to us. Specifically, we could first fit the data to c-shaped curvatures, extract the curvature values, and make our validation jigs incorporating the top 5 most commonly observed needle curvatures that fall within $< 2.0\ m^{-1}$ range (so as to not permanently deform the needle). We could perform a similar process and fit the clinical data onto s-shaped curvatures. This would further help us evaluate our shape sensing capabilities when faced with curvatures representative of real needle trajectory paths.

4.3.2 Stereo Camera Validation

Despite the effectiveness of our current stereo reconstruction pipeline, depth estimation from traditional stereo cameras remains prone to noise and limited resolution, especially in low-texture or homogeneous media like gel phantoms. To address this limitation, we plan to incorporate FoundationStereo — a state-of-the-art learning-based stereo reconstruction framework — into our pipeline. FoundationStereo leverages deep neural priors and pre-trained models to produce dense, high-fidelity depth maps even under challenging visual conditions. By applying this method to our experimental data, we aim to obtain smoother and more accurate 3D reconstructions of the inserted needle geometry. Once integrated, the FoundationStereo-based reconstructions will be quantitatively and qualitatively compared against shape reconstructions derived from Fiber Bragg Grating (FBG)-based needles. This cross-modality comparison will allow us to evaluate the spatial accuracy and robustness of stereo-based reconstruction methods in soft tissue environments, and potentially inform hybrid sensing strategies that fuse visual and FBG data.

4.3.3 Gel Phantom Experiment

To more closely mimic the heterogeneous structure of human soft tissue, we plan to develop and test multi-layer gel phantoms in future experiments. Unlike the current homogeneous phantom, the layered design will incorporate materials of varying stiffness, optical properties, and mechanical responses, simulating different anatomical regions. Then, testing the robustness of stereo reconstruction under complex visual conditions, where changes in optical scattering and refraction may occur at layer boundaries.
Furthermore, we also will improve the gel phantom cooking process. Since different transparency of gel phantoms can lead the differences in the calibration of the stereo camera and the validation of the stereo camera, we will improve the cooking procedure to ensure that every gel phantom has similar transparency.

4.3.4 Real Tissue Experiment

The final stage of the stylet reconstruction validation process will be to perform an experiment in the Mock OR inserting the stylet into real tissue phantoms with a similar consistency to cervical tissue. CombiCT will be used to reconstruct the stylet shape, in a similar manner to the stereo camera reconstruction.

4.4 Lessons Learned

When working on this project we learned some valuable lessons, notably:

1. Perform testing of external software being used. We had many issues due to not thoroughly understanding the FemtoSense software calculations. We believe the peak calculations used may not be optimal, so we plan to validate the software and replace it if necessary.

2. Check all measurements before 3D printing. We had to reprint jigs due to not fitting the tubing properly and needle holders due to the center point changing during rotations.

3. Be careful when applying force to plastic pieces. We accidentally pushed a plastic clamp to hard while trying to secure our jig and it snapped.

4.5 Credits

Yujie Zhang: Interrogator setup, stereo camera reconstruction, gel phantom production.
Siddhi Date: Jig and holder designs, 3D printing.
Kayleigh Huk: FBG calibration and reconstruction software.

Chapter 5

Technical Appendices

5.1 Required Documents Accessible from Wiki Page

Our Wiki page is linked here: CIS II Sp25 Group 6. The Wiki contains the following key documents:

1. Functional Specifications: Contains an overview of the entire system, as well as the necessary requirements for the stylet reconstruction software to be defined as a success.
2. Design Specifications: Contains an overview of the parts of our system architecture, including impementation requirements.
3. Testing Plan: Contains our validation plan for the software and hardware of our system.
4. User Guide: Contains an explanation of how to use our reconstruction software.
5. Calibration Procedure: Detailed steps to reproduce our calibration and validation experiments.

5.2 Software Repository

Github Repository: Our software is in the 'real_time_reconstruction' folder. The calibration dataset for the minimum deliverable is in the 'cal_dataset' folder.